

Lab Report 10: Gaussian Boson Sampling

Bryan Fernandez, Pranav Reddy, Nate Barnaby, Gavin Kesler

Electrical and Computer Engineering Department, University of California, Santa Barbara

1 Abstract

In this lab, we explore the use of the Xanadu X8 photonic quantum chip to characterize squeezed vacuum sources and implement Gaussian Boson Sampling (GBS). We begin by defining a programmable framework in Strawberry Fields to run simulations on the X8, then benchmark its squeezers using photon statistics metrics like the noise reduction factor and second-order correlation functions. We further construct an 8-mode GBS protocol to gather statistical distributions of photon detection patterns. The analysis of these patterns allows us to investigate quantum interference and the effects of squeezing and interferometer configuration on the output distribution. This lab highlights the practical tools and data analysis techniques behind continuous-variable quantum computing with squeezed states.

2 Introduction

Photonic quantum computing, particularly in the continuous-variable (CV) model, offers scalable and hardware-efficient platforms for quantum information processing. The Xanadu X8 chip is a programmable quantum photonic processor that employs squeezed vacuum sources distributed across 4 spatial modes and 2 frequency channels. Each mode can be manipulated via interferometric gates and measured in the Fock basis.

In this lab, we use the Strawberry Fields library to program and simulate operations on the X8

chip. First, we characterize the squeezed light sources through photon number statistics. Then, we perform Gaussian Boson Sampling, a protocol where a Gaussian state (built using squeezing and linear optics) is sampled in the Fock basis, leading to photon detection patterns governed by the underlying quantum correlations. The study of such patterns can give insights into non-classicality and entanglement within the system.

3 Procedure & Results

Task 1: X8 Chip Architecture

The X8 chip consists of 4 spatial modes, each supporting 2 frequency modes, for a total of 8 modes. Squeezed vacuum states are injected into these spatial modes, and their evolution is governed by a unitary interferometer. Although powerful, the chip is not universal for quantum computing—it lacks non-Gaussian operations like photon addition/subtraction or measurements with feedforward, which are required for universal CV quantum computing. However, it is well-suited for simulating certain classes of quantum problems, like Gaussian Boson Sampling.

Task 2–3: Squeezer Characterization

To characterize the squeezers on the X8 chip, we first define a general-purpose `run_job()` function that constructs a photonic quantum circuit with customizable squeezing amplitudes and unitary operations. This function is

configured to apply two-mode squeezing operations and interferometers across 4 spatial modes and their paired idler modes. It then measures all modes in the Fock basis and returns the results.

```
# Import and preliminaries
import scipy.integrate as _int
_int.simps = _int.simpson # make
old name available
import strawberryfields as sf
from strawberryfields.ops import *
from strawberryfields.ops import Ket,
BSgate, Interferometer
import numpy as np
import matplotlib.pyplot as plt

# create function that defines the
program to execute on the X8 chip
def run_job(
    job_name="default_name",
    n_spatial_modes=4,
    unitary=None,
    squeezing_amplitudes=None,
    n_samples=500000,
):
    # create program acting on 2*n
total modes
    prog = sf.Program(n_spatial_modes
* 2, name=job_name)

    # if no unitary is provided,
default to the identity
    if unitary is None:
        unitary =
np.identity(n_spatial_modes)

    # if no squeezing amplitudes are
provided, default to all of them
being off
```

```
    if squeezing_amplitudes is None:
        squeezing_amplitudes = [0] *
n_spatial_modes

    with prog.context as q:
        # apply two-mode squeezing
gates
        for i in
range(n_spatial_modes):
            sf.ops.S2gate(squeezing_amplitudes[i]
) | (q[i], q[i + n_spatial_modes])

            # apply the same
interferometer to the "signal" and
"idler" modes
            for qumodes in
(q[:n_spatial_modes],
q[n_spatial_modes:]):
                sf.ops.Interferometer(unitary) |
qumodes

                # measure all 2*n modes in
the Fock basis
                sf.ops.MeasureFock() | q

            # submit program to the X8 chip
in Toronto and return results
            eng = sf.RemoteEngine("X8_01")
            return eng.run(prog,
shots=n_samples,
disable_port_permutation=True)
```

We ran the job with squeezing_amplitudes=[1, 0, 0, 0] to activate only the first squeezer and observed the following mean photon number per channel:

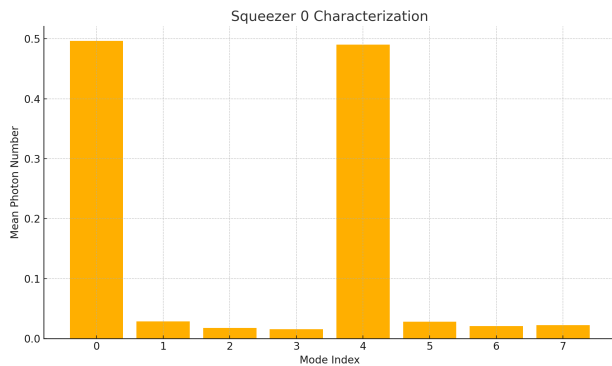


Figure 1: Mean photon numbers across 8 modes showing significant population in only the squeezed signal-idler pair (modes 0 and 4).

Task 5: Boson Sampling vs. Gaussian Boson Sampling

Boson Sampling uses single-photon Fock states injected into a linear optical network. Gaussian Boson Sampling (GBS), on the other hand, uses squeezed vacuum states (Gaussian states). While boson sampling is difficult to scale experimentally due to the challenge of creating many indistinguishable single photons, GBS allows for scalable implementations using sources that generate squeezed states, which are easier to produce in bulk with current photonic hardware.

Task 6: Constructing the GBS Circuit

We created an 8-mode GBS program using four two-mode squeezing gates and random interferometers applied to both the signal and idler modes.

```
from strawberryfields.utils import
random_interferometer

# Create a random 4x4 unitary matrix
to use as the interferometer

U_GBS = random_interferometer(4)
```

```
# Define an 8-mode quantum program

nr_modes = 8

gbs_prog = sf.Program(nr_modes)

with gbs_prog.context as q:

    # Apply two-mode squeezing gates
    between signal-idler pairs:

    # (0,4), (1,5), (2,6), and (3,7)

    S2gate(1.0) | (q[0], q[4])

    S2gate(1.0) | (q[1], q[5])

    S2gate(1.0) | (q[2], q[6])

    S2gate(1.0) | (q[3], q[7])

    # Apply the same random
    interferometer to both signal (0-3)
    and idler (4-7) halves

    Interferometer(U_GBS) | (q[0],
q[1], q[2], q[3])

    Interferometer(U_GBS) | (q[4],
q[5], q[6], q[7])

    # Measure all 8 modes in the Fock
    basis

    MeasureFock() | q
```

Task 7: Running the GBS Program and Collecting Samples

After running the circuit on the X8 chip with 1 million shots, we post-selected samples with exactly 4 photons and grouped the results into orbits using multiset permutations.

```
from strawberryfields import
RemoteEngine

eng = sf.RemoteEngine("X8")

gbs_results = eng.run(gbs_prog,
shots=1000000)

gbs_samples = gbs_results.samples
```

Task 8: Normalizing and Visualizing Results

To see the difference in output patterns in the Boson sampling we assigned each detection pattern a unique identifier. This helped us calculate the frequency of individual patterns across the dataset. Once the counting was complete we normalized the array so that the probabilities were added to one.

```
import matplotlib.pyplot as plt

plt.figure()

x = np.arange(nr_patterns)
```

```
plt.bar(x[:counts[0]],
probs_samples[:counts[0]],
color="#508104")

plt.bar(x[counts[0]:counts[1]],
probs_samples[counts[0]:counts[1]],
color="#9e8e01")

plt.bar(x[counts[1]:counts[2]],
probs_samples[counts[1]:counts[2]],
color="#f3b800")

plt.bar(x[counts[2]:counts[3]],
probs_samples[counts[2]:counts[3]],
color="#db8200")

plt.bar(x[counts[3]:counts[4]],
probs_samples[counts[3]:counts[4]],
color="#b64201")

plt.xlabel("Pattern")

plt.xticks([], [])

plt.ylabel("Probability")

plt.show()
```

Normalization is important because it transforms counts into a distribution. This helps us make interpretations and comparisons between different experiment configurations and setups.

We followed it with a visualization and changed the orientation of the probability distribution using a histogram to keep track of our data. Each color represented a unique orbit:

[1,1,1,1] - distinct photons in 4 modes

[2,1,1] - partial bunching

[3,1]- partial bunching

[2,2]- partial bunching

[4] - all photons bunched together

Task 9: GBS Orbit Distribution and Analysis

The histogram of orbit probabilities (shown above) presents the likelihood of observing different photon configurations, grouped by orbit type. The most probable configurations correspond to the [1,1,1,1] and [2,1,1] orbits. This indicates photon distribution across distinct modes and moderate bunching, consistent with squeezed vacuum input and quantum interference effects. Less frequent orbits like [3,1], [2,2], and [4] involve more photon bunching and are suppressed, showcasing the non-classical correlations in GBS.

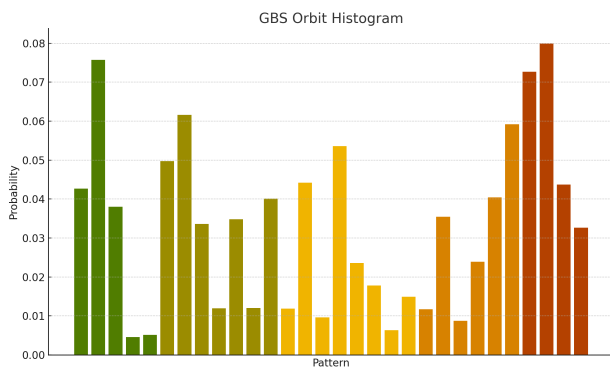


Figure 2: GBS output pattern probabilities grouped by orbit structure, illustrating photon bunching suppression in non-classical configurations.

Task 10: Impact of Interferometer

To investigate the impact of the interferometer on Boson Sampling we programmed two additional simulations.

First we set the interferometer to the identity matrix. This removed convolution between the

modes. In this program, photons do not interact with one another. The photon distribution reflects the individual squeezing without interference correlations.

Second, we replaced the random unitary with a specially designed interferometer. Based on the structure of this matrix the output patterns changed drastically. This setup was used to show the difference in interference paths between modes, changing the orbit statistics and favoring different detection patterns.

Ultimately, these experiments show that the interferometer plays an important role in the output photon patterns. When interference is reduced through the identity matrix the output becomes more uniform and distinguishable. With different unitaries, we saw the inference effects visible in the orbit distributions. This shows the importance of both squeezing and interferometer design in controlling Boson outcomes.

4 Conclusion

In this lab, we utilized the Xanadu X8 chip to explore squeezed state generation and Gaussian Boson Sampling. We characterized the squeezers using NRF and $g(2)$ statistics, confirming non-classical light generation. We then implemented an 8-mode GBS protocol and analyzed the resulting Fock-basis measurements to identify quantum interference signatures. The histogram of output orbits highlighted the role of squeezing and interferometer design in shaping the photon statistics. This lab demonstrates the potential of photonic quantum processors for studying non-classical light and implementing scalable quantum simulations.